SDS

Day-5

→ Perceptron Trick

- In this trick. we take a line randomly and run a loop (eg 1000) and ask every point that they classified correctly if yes then no changes if not correctly then we perform task we push / pull the line according to the point y axis

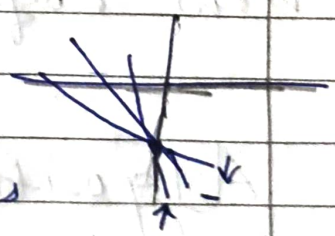
→ How to level Region?

if

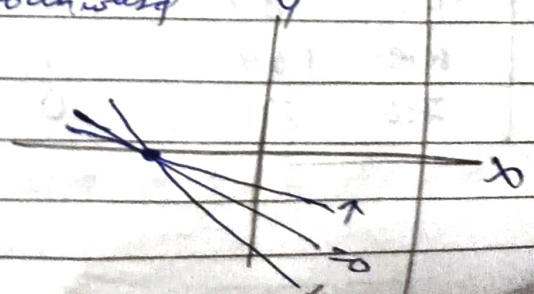
$Ax+By+C = 0$ → on the line point present
$Ax+By+C > 0$ → on the positive side
$Ax+By+C < 0$ → on the Negative Region.

- if we do changes in C then it moves parallely according to sign.
  C ↑ downward C ↓ upward

- Changes in x then it moves around x axis where y axis term is fixed
  x ↑ left move x ↓ Right move.



- Change in y then it moves around y axis where x axis term is fixed
  y ↑ upward y ↓ downward



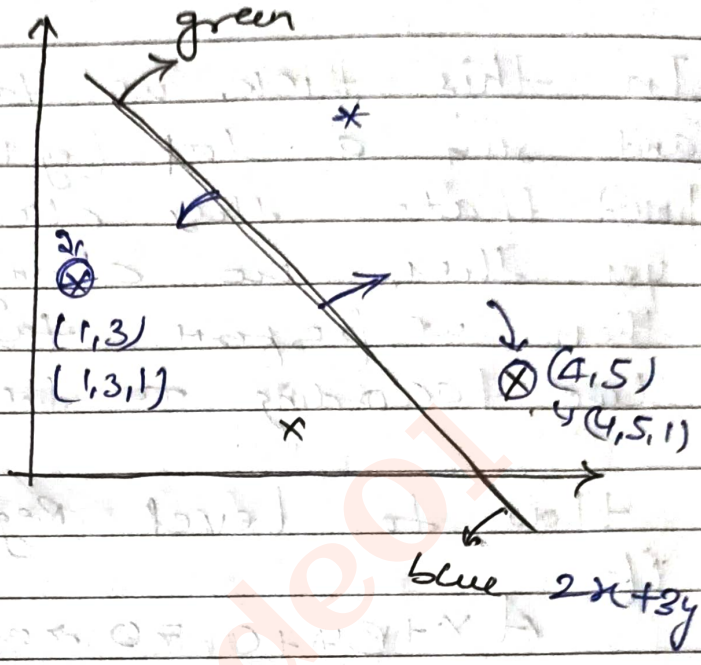
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→ Transformation

2 3 5
-4 5 1
-2 -2 +4 -
$-2x-2y+4=0$

2 3 5
+1 3 1
3 0 6



negative point in positive Region { if we want to move in negative dir. then we add the coodinate

positive point in the negative Region { if we want to move line in positive region the we substract the coordinate

→ Learning rate(η) → 0.01

new coef = old coef - η coordinate of point

→ Algorithm of perception for code

x_0	(x_1) cgpa	(x_2) iq	y placed
1	7.5	81	1
1	8.9	109	1
1	7.0	81	0

$Ax+By+C=0$

$w_0+w_1x_1+w_2x_2=0$

$w_0=C,\ w_1=A,\ w_2=B$

$w_0x_0+w_1x_1+w_2x_2=0$ (this)

SDS

$\sum_{i=0}^{n} w_i x_i = 0$ → General eq for line

prediction for 1

$w_0 \cdot 1 + w_1 \times 7.5 + w_2 \cdot 81 = \underline{\quad} \geq 0$ 1 Y
< 0 0 N

$$[w_0 \; w_1 \; w_2] \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$$

→ epoch value → 1000 , η (learning rate) = 0.01

for i in range (epochs):
randomly select a student

negative point → positive Region { if $x_i \in N$ and $\sum_{i=0}^{2} w_i x_i \geq 0$ (+ve)
$w_{new} = w_{old} - \eta x_i$

positive point → negative region { if $x_i \in P$ and $\sum_{i=0}^{2} w_i x_i < 0$
$w_{new} = w_{old} + \eta x_i$

→ Simplified Algorithm

$x_i \in N$ and $\sum_i w_i x_i \geq 0$
$w_n = w_o - \eta x_i$

$x_i \in P$ and $\sum_i w_i x_i < 0$
$w_n = w_o + \eta x_i$

→ $w_n = w_o + \eta (y_i - \hat{y}_i) \cdot x_i$

shivamxcodes